

AI503- Machine Learning
Lecture 3: Supervised Learning – Decision
Tree Classification

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Machine Learning Classification

- Given a collection of records (training set)
- Each record = set of attributes, one of attributes = class.
- Find a model for class attribute as a function of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
- A test set is used to determine the accuracy of the model. More on that later

Decision Tree

- Decision trees are a simple machine learning tool used for classification and regression tasks. They break complex decisions into smaller steps, making them easy to understand and implement.

Understanding Decision Tree

- A decision tree, which has a hierarchical structure made up of root, branches, internal, and leaf nodes, is a non-parametric supervised learning approach used for classification and regression applications.
- It is a tool that has applications spanning several different areas. These trees can be used for classification as well as regression problems. The name itself suggests that it uses a flowchart like a tree structure to show the predictions that result from a series of feature-based splits. It starts with a root node and ends with a decision made by leaves.

Working on Decision Tree Algorithm

- In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree, and the decision tree algorithm compares the values of the root attribute with the record (real dataset) attribute and, based on the comparison, follows the branch and jumps to the next node. The algorithm again compares the attribute value with the other sub-nodes for the next node and moves further. It continues the process until it reaches the leaf node of the tree.

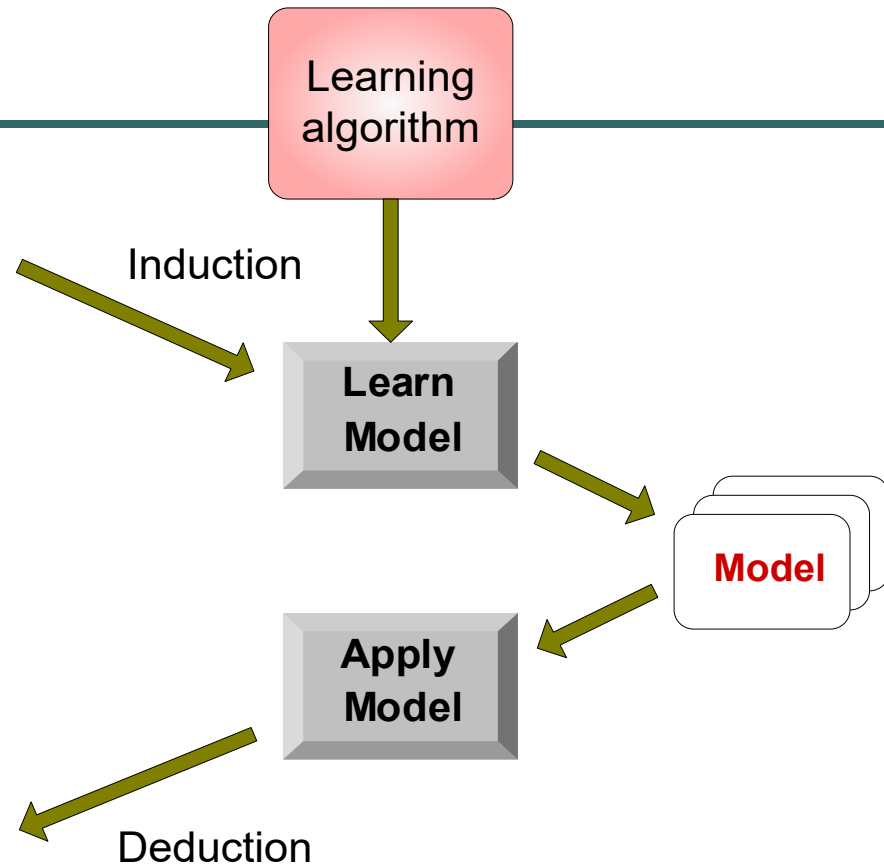
Illustrating Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

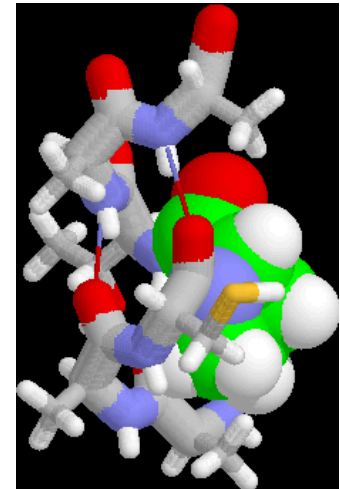
Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Examples of Classification Task

- ❑ **Predicting tumor cells as benign or malignant**
- ❑ **Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil**
- ❑ **Categorizing news stories as finance, weather, entertainment, sports, etc**



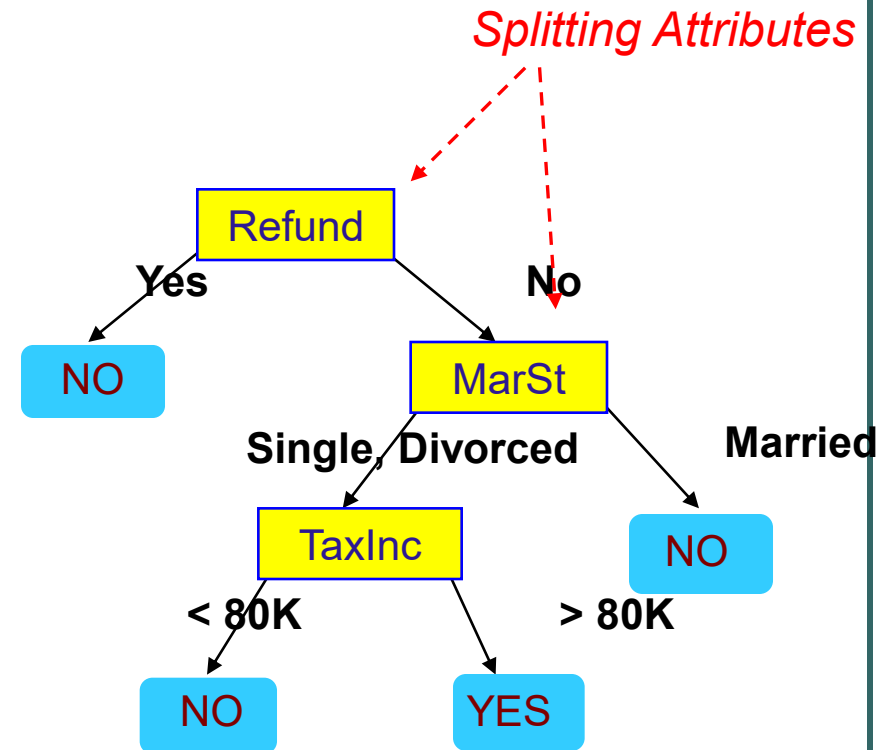
Working on Decision Tree Algorithm

- The complete process can be better understood using the below algorithm:
- **Step-1: Select** the root node based on the information gained value from the complete dataset.
- **Step-2:** Divide the root node into sub-nodes based on information gain and entropy values.
- **Step-3:** Continue this process till we cannot further classify the node into sub-nodes called leaf nodes.

Example of a Decision Tree

<i>Tid</i>	<i>Refund</i>	<i>Marital Status</i>	<i>Taxable Income</i>	<i>Cheat</i>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training Data

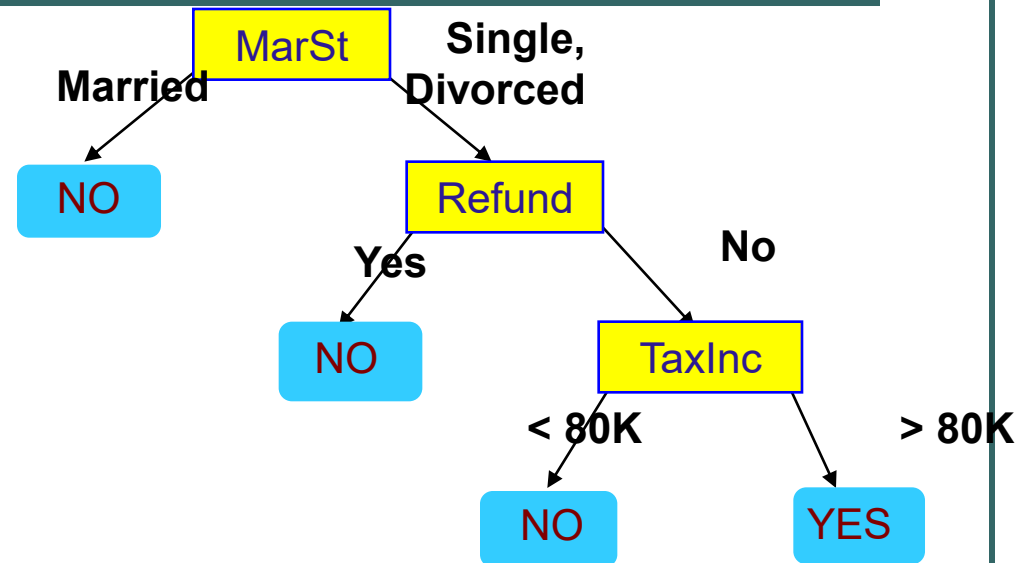


Model: Decision Tree

Another Example of Decision Tree

categorical categorical continuous class

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

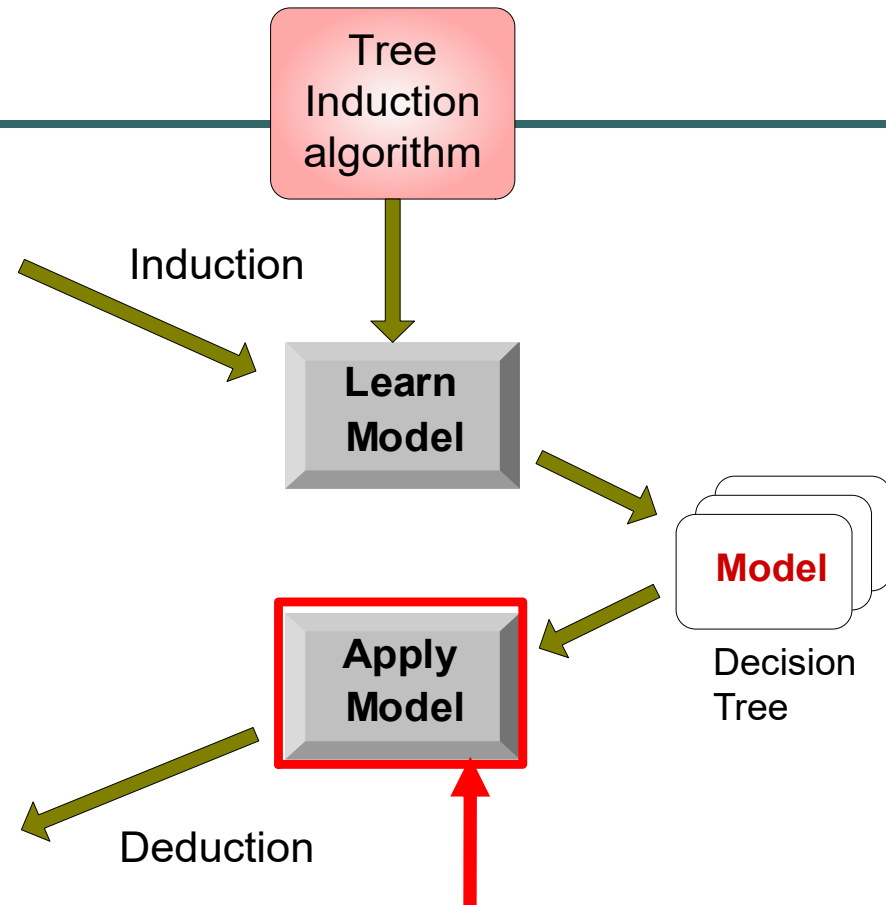
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
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6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

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Test Set

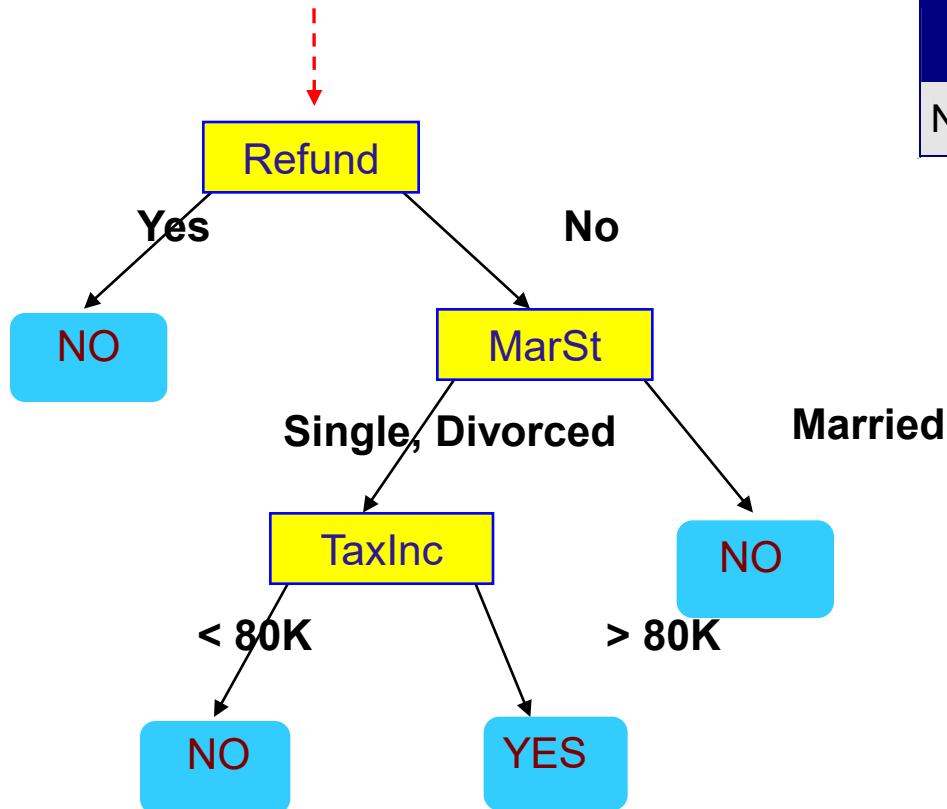


Apply Model to Test Data

Test Data

Start from the root of tree.

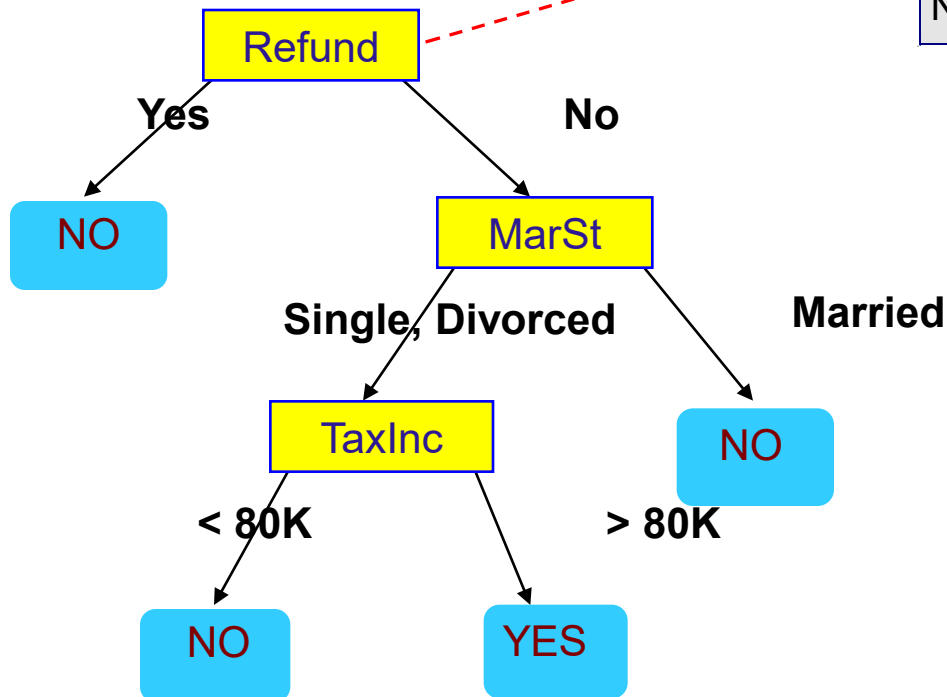
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

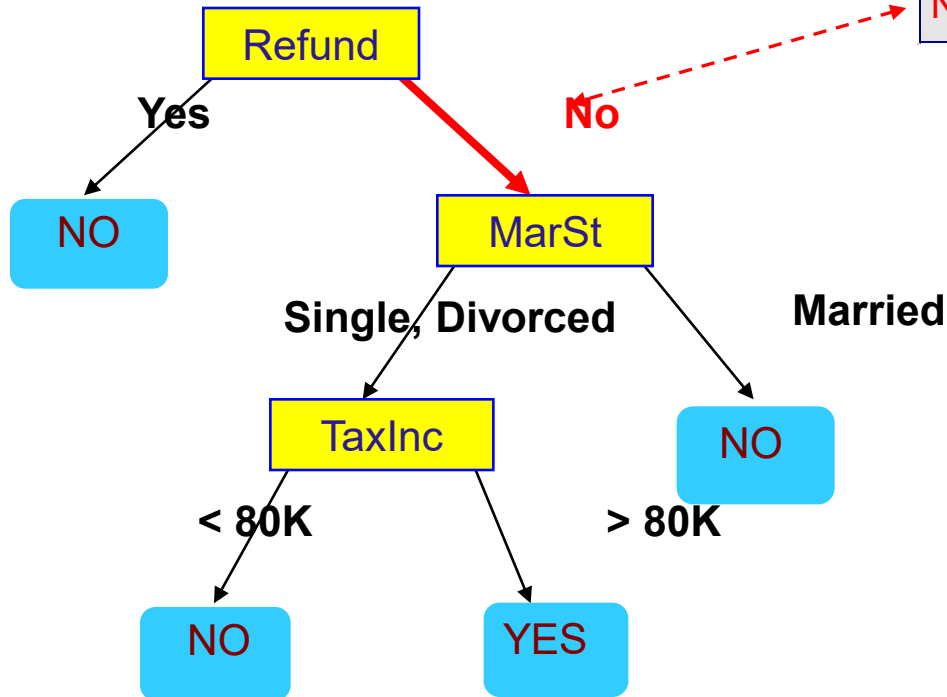
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

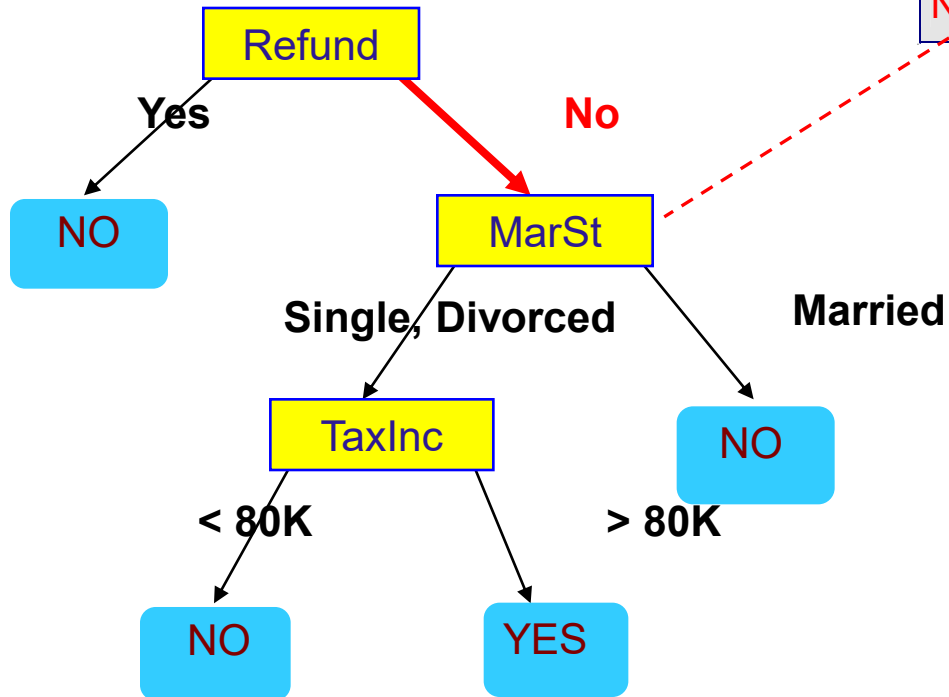
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

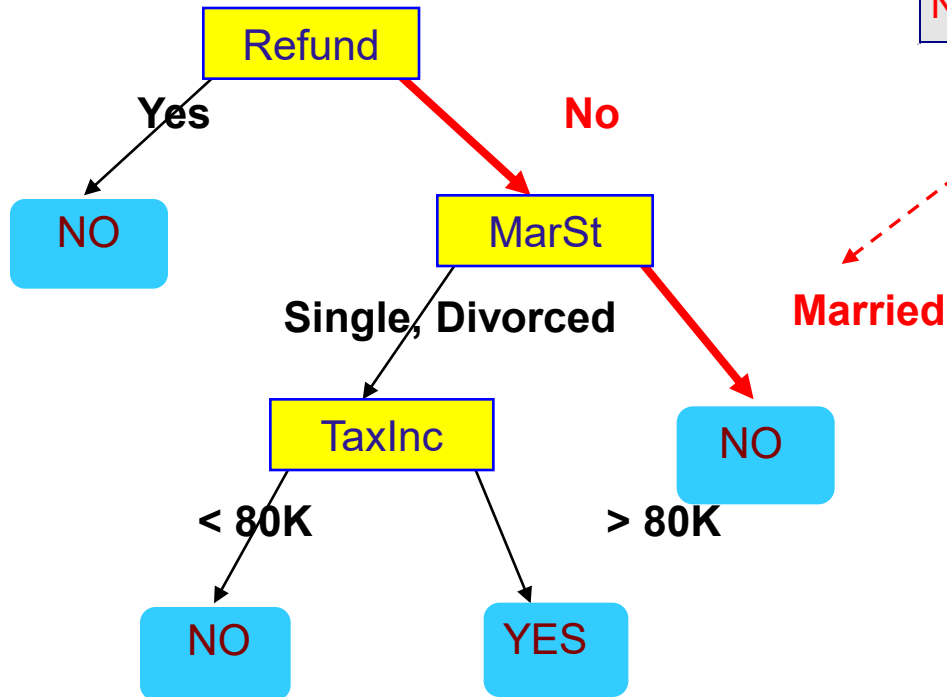
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

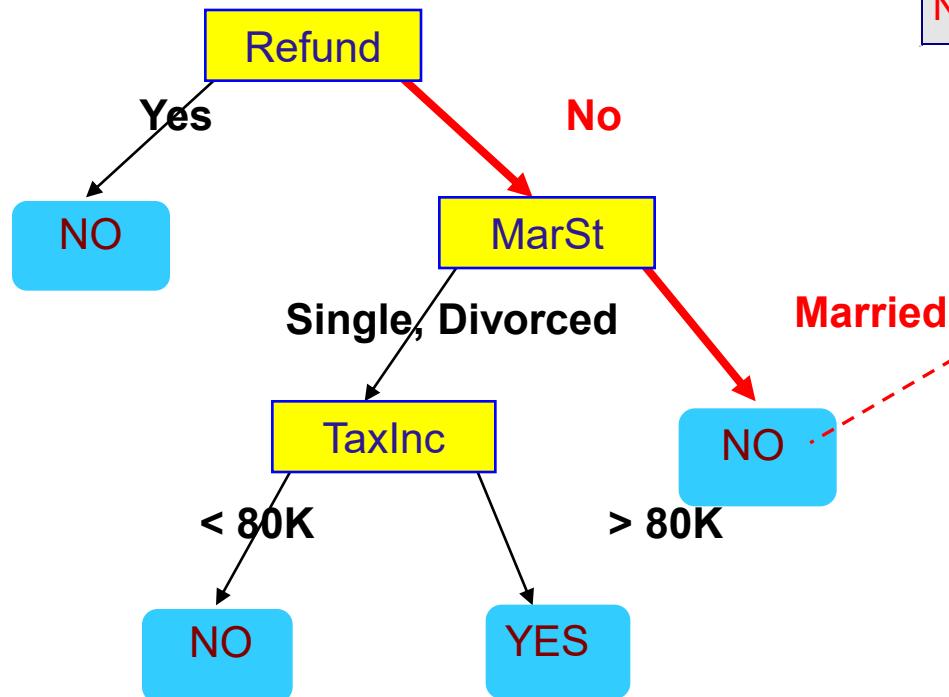
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Assign Cheat to
"No"

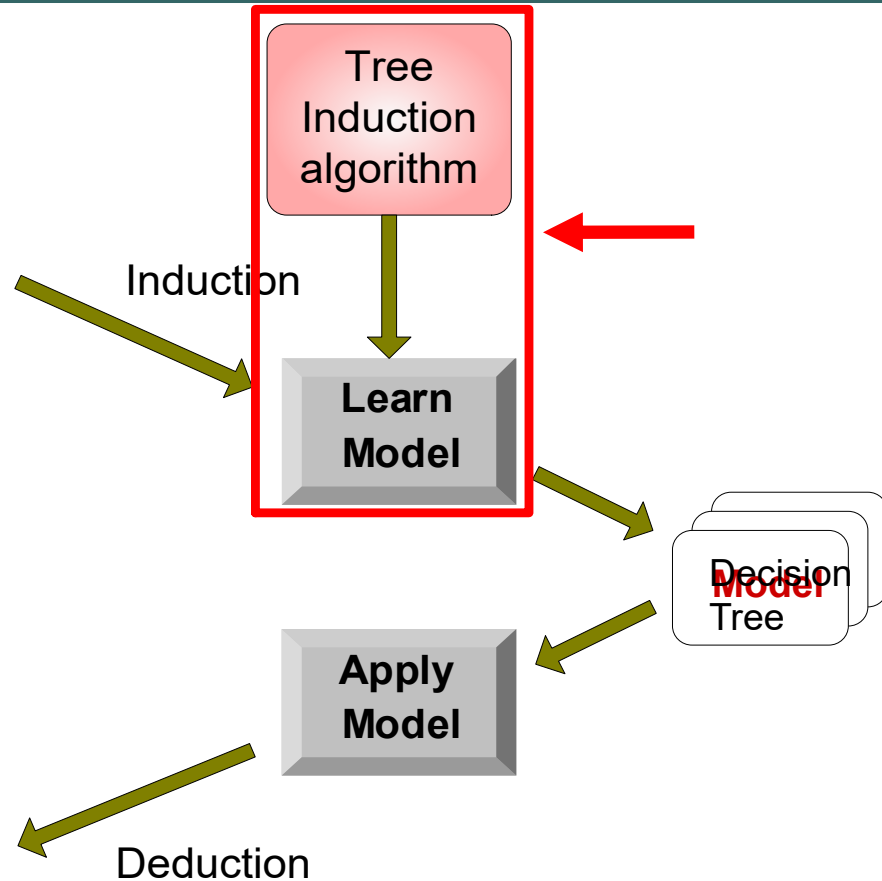
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Test Set



Decision Tree Terminologies

Before learning more about decision trees let's get familiar with some of the terminologies:

- **Root Node:** The initial node at the beginning of a decision tree, where the entire population or dataset starts dividing based on various features or conditions.
- **Decision Nodes:** Nodes resulting from the splitting of root nodes are known as decision nodes. These nodes represent intermediate decisions or conditions within the tree.
- **Leaf Nodes:** Nodes where further splitting is not possible, often indicating the final classification or outcome. Leaf nodes are also referred to as terminal nodes.

Decision Tree Terminologies

- ***Sub-Tree***: Similar to a subsection of a graph being called a sub-graph, a sub-section of a these tree is referred to as a sub-tree. It represents a specific portion of the decision tree.
- ***Pruning***: The process of removing or cutting down specific nodes in a tree to prevent overfitting and simplify the model.



Decision Tree Terminologies

- ***Branch / Sub-Tree:*** A subsection of the entire is referred to as a branch or sub-tree. It represents a specific path of decisions and outcomes within the tree.
- ***Parent and Child Node:*** In a decision tree, a node that is divided into sub-nodes is known as a parent node, and the sub-nodes emerging from it are referred to as child nodes. The parent node represents a decision or condition, while the child nodes represent the potential outcomes or further decisions based on that condition.

